

Bayesian Monitoring

Posterior probability threshold
 γ^* calibrated at a single point

✓ PASS

Blinded SSR

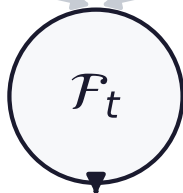
Conditional-power rule
at interim look $n = 45$

✓ PASS

Response-Adaptive Randomization

Thompson Sampling
allocation

✓ PASS



Shared interim information set

All three rules read the same
interim data at the same look

Composed Decision Pipeline

$P_0(\phi_{\text{comp}}(X) = 1)$ is not constrained by the component checks

Calibrated at one point

$p_0 = 0.25$, no trend
Type I error = 0.0250



Evaluated across the composite null

Type I error 0.0124 at $p = 0.20$ to 0.2618 at $p = 0.36$
(shared time trend active; nominal 0.025)